Kirinyaga University

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SCHOOL: [Insert School]
DEPARTMENT: [Insert Department]
PROGRAMME: [Insert Programme]
YEAR: [Insert Year] **SEMESTER:** [Insert Semester]
UNIT CODE: COM 312 **UNIT TITLE:** System Programming
LECTURE HOURS: 42 (3 hrs/week) **PRE-REQUISITES:** [Insert Pre-requisites]
LECTURER'S NAME: Mr. Samuel Maina
LECTURER'S CONTACTS: TEL: [Insert Tel]
EMAIL: samicho4@gmail.com

Welcome Message

Welcome to COM312: System Programming. This course introduces you to programming operating-system modules, giving you a working command of process control, device and signal handling, memory and file management, and inter-process/inter-machine communication using C/C++.

Learning Outcomes

At the end of the course the learner will be able to:

- Explain the scope and tasks of systems programming
- Describe processes as fundamental objects in the OS
- Describe the basic concepts of threads and multithreaded programming
- Describe system calls for files and use programming operations for files and directories

Teaching Methodology

- Stand-up instruction (lectures)
- Presentations and guided discussions
- Tutorials
- Individual and group exercises
- Practical-oriented exercises

Instructional Materials

- Whiteboards
- LCD projectors
- Computers
- Resource persons

Course Evaluation

- CAT 1 = 10% | CAT 2 = 10% | Assignments = 10%
- End of Semester Examination = 70% (Total = 100%)
- Pass mark: 40%

COURSE OUTLINE

S/N.	Week	Sub-Topic	Remarks
1.	**An Introduction to Systems Programming and Essential Concepts**	• Design principles of Linux • Overview of systems programming and objectives • Components of systems programming • Theory and practical of system calls using C	3 hrs
2.	**Preparing Your Tool Box**	• GNU gcc and gdb • NASM (Netwide Assembler) • UNIX readelf, objdump, od, nm, ldd and ld config • Creating your own static and dynamic libraries	3 hrs
3.	**UNIX File System**	• Linux directory structures • Basic shell scripting • UNIX and POSIX file systems • Inodes, APIs, and UNIX kernel support for files	3 hrs
4.	**CAT I**	• Contents covered	3 hrs
5.	**Process Management and Scheduling**	• Process identification • Process trees, chains and fans • Zombie and orphan processes • Process groups, sessions and controlling terminals • UNIX daemons • System V R3 and System V R4 schedulers	3 hrs
6.	**UNIX Inter-Process Communication Tools**	• Signals • Pipes / FIFOs • Message queues • Shared memory • Memory mapping	3 hrs
7.	**Thread Management and Synchronization**	• POSIX thread API • Introduction to threads • Synchronization among threads using mutexes and condition variables • Synchronization among related and unrelated processes using semaphores	3 hrs

S/N.	Week	Sub-Topic	Remarks
8.	**Network Programming**	• UNIX socket API • Writing concurrent servers – multi-process model • Writing concurrent servers – multi-threaded model • Non-blocking I/O using the select() system call	3 hrs
9.	**Network Security**	• Buffer overflow vulnerabilities • Exploiting vulnerabilities using Metasploit frameworks	3 hrs
10.	**Group Presentation Practicals (Assignment)**	• Identified area for group presentation: pipes practical	3 hrs
11.	**Group Presentation Practicals (Assignment)**	• Identified area for group presentation: network programming	
12.	**CAT 2**	• Contents covered	3 hrs
13.	**Linux User Management (Personal Reading Topic)**	• Linux users • Linux user management • User passwords • Local groups	3 hrs
14.	**Revision Week**	• Course revisions	3 hrs

COURSE TEXTBOOK

i) Stevens, W. R. Advanced Programming in the UNIX Environment.

REFERENCE TEXTBOOKS

i) Kerrisk, M. (2010). The Linux Programming Interface: A Linux and UNIX System Programming Handbook. No Starch Press.

ii) Raymond, E. S. The Art of UNIX Programming.